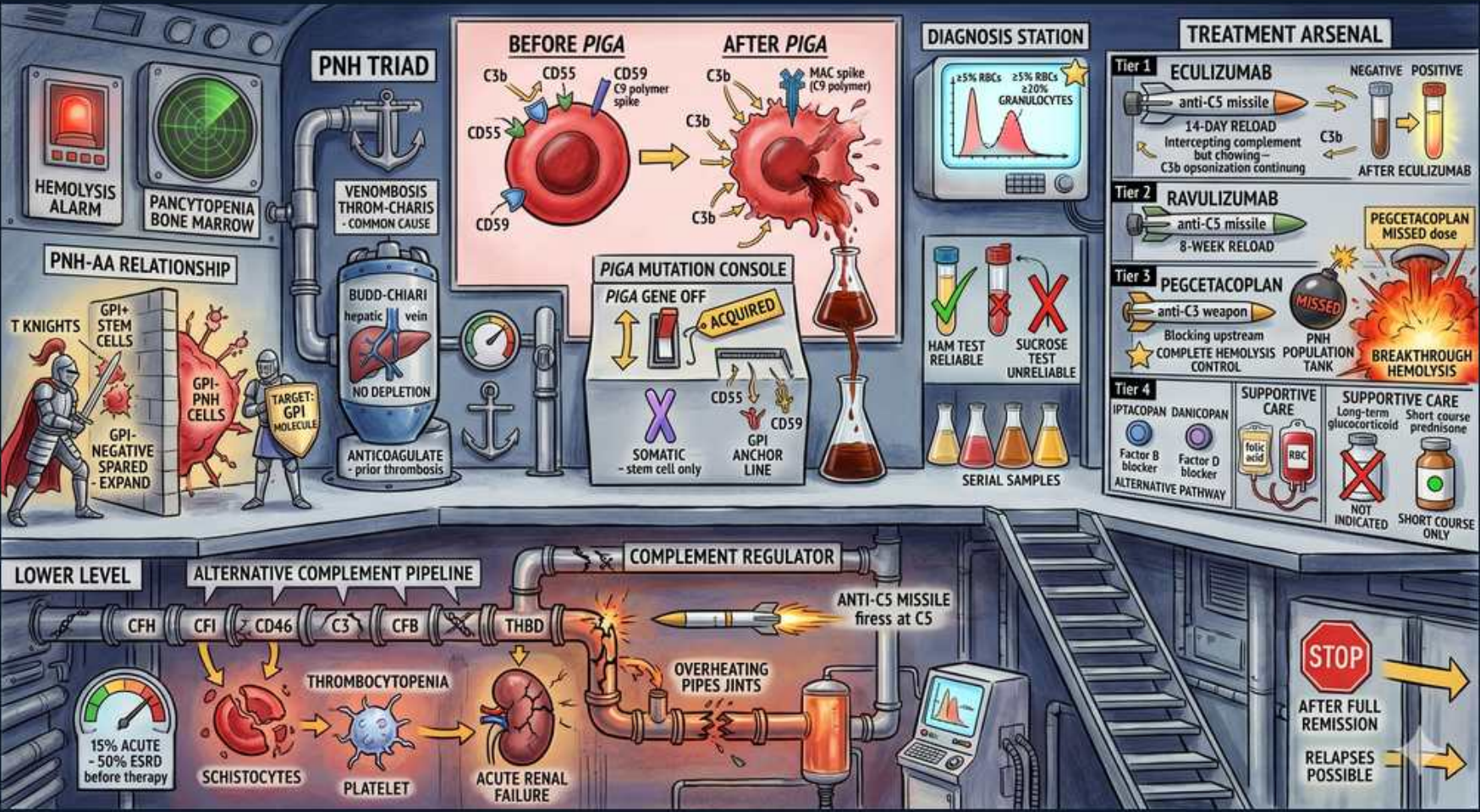


Paroxysmal Nocturnal Hemoglobinuria — PIGA, Complement & Anti-C5 Therapy



1. PIGA mutation — somatic, X-linked

WHAT YOU SEE

Mutation console: PIGA gene acquired

WHAT IT ENCODES

PNH arises from an ACQUIRED SOMATIC mutation in PIGA in a hematopoietic stem cell. PIGA makes GPI anchors; without them, RBCs lose the complement regulators CD55 and CD59, becoming hypersensitive to complement lysis.

BOARD TRAP

PNH is acquired/somatic, NOT inherited; PIGA is X-linked so one hit suffices in males' single active X — but the mutation is clonal, not germline.

2. CD55 / CD59 anchor loss

WHAT YOU SEE

Before/after RBC missing CD55/CD59

WHAT IT ENCODES

GPI-anchored CD55 (DAF) accelerates C3 convertase decay; CD59 blocks MAC (C5b-9) formation. Loss of BOTH leaves RBCs defenseless against complement, causing chronic INTRAVASCULAR hemolysis.

BOARD TRAP

CD59 loss (MAC) is the dominant driver of intravascular hemolysis; flow cytometry for CD55/CD59 deficiency (FLAER) is the diagnostic test, not the Ham/sucrose tests.

3. Intravascular hemolysis — hemoglobinuria

WHAT YOU SEE

RBC lysing, dark urine

WHAT IT ENCODES

Complement-mediated intravascular lysis releases free hemoglobin, scavenging nitric oxide (causing dystonia, esophageal spasm, fatigue, pulmonary hypertension) and producing dark morning urine.

BOARD TRAP

PNH hemoglobinuria is intravascular (low haptoglobin, high LDH, hemoglobinuria), contrasting extravascular AIHA; the NO-scavenging symptoms are tested.

4. Venous thrombosis — Budd-Chiari

WHAT YOU SEE

Thrombus, Budd-Chiari common cause

WHAT IT ENCODES

THROMBOSIS is the leading cause of death in PNH, classically at unusual sites: hepatic vein (Budd-Chiari), cerebral, and abdominal veins. Free Hb, platelet activation, and complement drive the prothrombotic state.

BOARD TRAP

Unexplained Budd-Chiari or splanchnic/cerebral vein thrombosis with hemolysis should prompt PNH testing — anticoagulation alone is insufficient without complement blockade.

5. Eculizumab — anti-C5

WHAT YOU SEE

Eculizumab missile at C5

WHAT IT ENCODES

Eculizumab is a monoclonal antibody against C5 that blocks MAC formation, reducing intravascular hemolysis, transfusion need, and thrombosis risk in PNH.

BOARD TRAP

Eculizumab controls INTRAVASCULAR hemolysis but can unmask C3-mediated EXTRAVASCULAR hemolysis (breakthrough) — explaining persistent anemia despite therapy.

6. Ravulizumab — 8-week dosing

WHAT YOU SEE

Ravulizumab vial

WHAT IT ENCODES

Ravulizumab is a long-acting anti-C5 antibody given every 8 weeks (vs eculizumab every 2 weeks), improving convenience with equivalent C5 blockade.

BOARD TRAP

Ravulizumab vs eculizumab differ mainly in dosing interval (8-weekly vs 2-weekly), not mechanism — both are anti-C5.

7. Pegcetacoplan — anti-C3

WHAT YOU SEE

Pegcetacoplan, breakthrough hemolysis

WHAT IT ENCODES

Pegcetacoplan blocks proximal complement at C3, addressing the C3-mediated extravascular hemolysis (breakthrough) that anti-C5 therapy leaves behind, raising hemoglobin further.

BOARD TRAP

If anemia persists on eculizumab due to C3 opsonization, switching to a C3 inhibitor (pegcetacoplan) targets the residual extravascular hemolysis.

8. Meningococcal vaccination required

WHAT YOU SEE

Vaccine/STOP sign

WHAT IT ENCODES

Complement inhibitors block terminal complement, sharply raising the risk of Neisseria meningitidis infection. Meningococcal vaccination (and often prophylactic antibiotics) is MANDATORY before starting anti-C5/C3 therapy.

BOARD TRAP

Failing to vaccinate against meningococcus before eculizumab is a board-classic safety error — terminal complement blockade predisposes to Neisseria.

9. Ham test / sucrose lysis (historical)

WHAT YOU SEE

Diagnosis station flasks

WHAT IT ENCODES

The acidified-serum (Ham) and sucrose lysis tests historically detected complement-sensitive RBCs in PNH but are now replaced by FLOW CYTOMETRY for GPI-anchored proteins (CD55/CD59, FLAER).

BOARD TRAP

Modern PNH diagnosis is flow cytometry (FLAER); Ham/sucrose tests are outdated trivia, less sensitive/specific.

10. PNH-AA relationship — aplastic anemia

WHAT YOU SEE

PNH-AA relationship panel

WHAT IT ENCODES

PNH frequently coexists with or evolves from aplastic anemia and myelodysplasia; the GPI-deficient clone expands under immune pressure that spares it. Check for PNH clones in AA/MDS.

BOARD TRAP

A PNH clone in aplastic anemia signals immune-mediated marrow failure; screening for it changes prognosis and therapy.

11. Pancytopenia / hypocellular marrow

WHAT YOU SEE

Hemolysis alarm + bone marrow

WHAT IT ENCODES

Because PNH stems from a marrow stem cell defect, patients often have associated bone marrow failure with pancytopenia, not just hemolysis. Reticulocyte response may be blunted.

BOARD TRAP

Pancytopenia in PNH reflects underlying marrow failure (overlap with AA) — anemia isn't solely from hemolysis.

12. Schistocytes / low-grade signs

WHAT YOU SEE

Lower panel: schistocytes, renal

WHAT IT ENCODES

Chronic intravascular hemolysis can produce renal injury (hemosiderin deposition, AKI) and other end-organ effects; LDH is markedly elevated and haptoglobin undetectable.

BOARD TRAP

Markedly high LDH with undetectable haptoglobin and hemoglobinuria signals intravascular hemolysis — the PNH lab signature.

13. Alternative complement pipeline regulators

WHAT YOU SEE

CFH/CFI/CD46 regulator pipeline

WHAT IT ENCODES

Complement regulatory proteins (CFH, CFI, CD46, CFB) restrain the alternative pathway; their balance underlies why GPI-anchor loss in PNH leads to uncontrolled complement attack on RBCs.

BOARD TRAP

PNH hemolysis is a defect in cell-surface regulators (CD55/CD59), distinct from soluble-regulator complementopathies — same cascade, different control point.

14. Acute renal failure from hemolysis

WHAT YOU SEE

Kidney, acute renal failure

WHAT IT ENCODES

Free hemoglobin and hemosiderin from chronic intravascular hemolysis can cause tubular injury and acute or chronic kidney disease in PNH.

BOARD TRAP

Renal impairment in PNH is hemolysis-driven (pigment nephropathy/hemosiderinosis), an end-organ complication beyond the hematologic picture.

15. Relapses possible after remission

WHAT YOU SEE

STOP sign: relapses after full remission

WHAT IT ENCODES

Rare spontaneous remissions occur, but PNH can relapse; therapy controls but does not cure. Allogeneic stem cell transplant is the only curative option, reserved for severe marrow failure.

BOARD TRAP

Complement inhibitors are disease-controlling, not curative; only allogeneic transplant cures PNH and is reserved for refractory/aplastic cases.
